

Cefazolin: Drug information - UpToDate

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For additional information see "Cefazolin: Patient drug information" and "Cefazolin: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Pharmacologic Category

- Antibiotic, Cephalosporin (First Generation)

Dosing: Adult

Dosage guidance:

Dosing: Dosing is presented based on the traditional intermittent infusion method over 30 to 60 minutes. Total daily dose may also be administered as a continuous infusion over 24 hours (Ref).

Bloodstream infection

Bloodstream infection:

Pathogen-directed therapy for methicillin-susceptible staphylococci:

IV: 2 g every 8 hours (Ref); treat uncomplicated *Staphylococcus aureus* bacteremia for ≥ 14 days starting from day of first negative blood culture, with longer courses warranted for endocarditis or metastatic sites of infection (Ref).

Pathogen-directed therapy for susceptible Enterobacteriaceae:

IV: 2 g every 8 hours (Ref). Usual duration is 7 to 14 days; individualize depending on source and extent of infection as well as clinical response. A 7-day duration is recommended for patients with uncomplicated Enterobacteriaceae infection who respond appropriately to antibiotic therapy (Ref).

Antibiotic lock technique (catheter-salvage strategy): Note: For infections caused by susceptible organisms when the catheter cannot be removed; use in addition to systemic antibiotics. Catheter salvage is **not** recommended for *S. aureus*.

Intracatheter: Prepare lock solution to final concentration of cefazolin 5 to 10 mg/mL (may be combined with heparin). Instill into each lumen of the catheter access port using a volume sufficient to fill the catheter (2 to 5 mL), with a dwell time of up to 72 hours depending on frequency of catheter use and solution stability. Withdraw lock solution prior to catheter use; replace with fresh cefazolin lock solution after catheter use. Antibiotic lock therapy is given for the same duration as systemic antibiotics (Ref).

Endocarditis, prophylaxis

Endocarditis, prophylaxis (dental or invasive respiratory tract procedures) (alternative agent) (off-label use):

Note: For patients with nonsevere, non-IgE-mediated penicillin allergy who cannot take oral therapy (Ref).

IM, IV: 1 g as a single dose 30 to 60 minutes before procedure; if inadvertently not given prior to the procedure, may be administered up to 2 hours after the procedure. **Note:** Reserve for select situations (cardiac condition with the highest risk of adverse endocarditis outcomes and procedure likely to result in bacteremia with an organism that can cause endocarditis) (Ref).

Endocarditis, treatment

Endocarditis, treatment (alternative agent):

Note: For pathogen-directed therapy for methicillin-susceptible staphylococci; reserve for patients with nonsevere, non-IgE-mediated penicillin allergy (Ref).

Native valve: **IV:** 2 g every 8 hours for 6 weeks. **Note:** For uncomplicated right-sided IE, 2 weeks of therapy may be adequate (Ref).

Prosthetic valve: **IV:** 2 g every 8 hours for ≥ 6 weeks (combine with rifampin for entire duration of therapy and gentamicin for the first 2 weeks) (Ref).

Intra-abdominal infection, mild to moderate, community acquired in patients without risk factors for resistance or treatment failure

Intra-abdominal infection, mild to moderate, community acquired in patients without risk factors for resistance or treatment failure:

Note: Reserve for patients with low risk for resistant pathogens (eg, local Enterobacteriaceae resistance rate to cefazolin $<10\%$ and no recent antibiotic exposure) (Ref).

Cholecystitis, acute: IV: 1 to 2 g every 8 hours; continue for 1 day after gallbladder removal or until clinical resolution in patients managed nonoperatively (Ref). **Note:** The addition of anaerobic therapy (eg, metronidazole) is recommended if biliary-enteric anastomosis is present (Ref).

Other intra-abdominal infection (eg, perforated appendix, appendiceal abscess, diverticulitis) (off-label use): IV: 1 to 2 g every 8 hours in combination with metronidazole. Total duration of therapy (which may include transition to oral antibiotics) is 4 to 5 days following adequate source control (Ref). For diverticulitis or uncomplicated appendicitis managed without intervention, duration varies from 4 to 14 days depending on disease extent, response, and immune status (Ref); for perforated appendicitis managed with laparoscopic appendectomy, 2 to 4 days may be sufficient (Ref).

Osteomyelitis and/or discitis

Osteomyelitis and/or discitis:

Treatment:

Note: For pathogen-directed therapy for methicillin-susceptible *S. aureus* (Ref).

IV: 2 g every 8 hours; duration of therapy is generally 6 weeks but depends on clinical response and presence of residual infection or hardware. Some patients who receive initial IV antibiotics can be transitioned to oral therapy to complete the course. Shorter courses are appropriate if the affected bone is completely resected by amputation (Ref).

Prevention, following open fractures:

IV: 2 g for patients <120 kg or 3 g for patients ≥ 120 kg every 8 hours; ideally administer within 6 hours of injury. For type I or II fractures (no more than moderate comminution or contamination, no or minimal periosteal stripping, adequate soft tissue coverage), discontinue 24 hours following wound closure. For type III fractures (severe contamination or comminution), use as part of an appropriate combination regimen and continue for 72 hours after injury or up to 24 hours after wound closure (Ref).

Note: For patients with risk for methicillin-resistant *S. aureus* (MRSA), potential water exposure, or fecal or clostridial contamination, alternative or additional antibiotics are recommended (Ref).

Peritonitis, treatment

Peritonitis, treatment, peritoneal dialysis (off-label use):

Note: Intraperitoneal administration is preferred to IV administration unless the patient has sepsis. Consider a 25% dose increase in patients with significant residual renal function (urine output >100 mL/day) (Ref).

Intermittent (long dwell): Intraperitoneal: 15 mg/kg added to the dialysate solution once daily; allow to dwell for ≥ 6 hours (Ref).

Intermittent (short dwell for automated peritoneal dialysis using on-warmer fluid draw): Intraperitoneal: 20 mg/kg added to the first 5 L on warmer dialysate solution once daily prior to rapid exchanges (ie, studied population utilized 2 L for 5 exchanges over a 10-hour treatment period) without a last fill or additional dwell (Ref).

Continuous (with every CAPD exchange): Intraperitoneal: Loading dose: 500 mg/L of dialysate added to first dialysate exchange; maintenance dose: 125 mg/L of dialysate with each subsequent dialysate exchange (Ref).

Duration: For patients with adequate clinical response, duration of therapy is ≥ 2 to 3 weeks depending on organism (Ref). For patients with no improvement after 5 days, remove catheter and treat with appropriate systemic antibiotics for 14 days after catheter removal (Ref).

Pneumonia

Pneumonia:

Note: For pathogen-directed therapy for methicillin-susceptible *S. aureus* (Ref).

IV: 2 g every 8 hours (Ref). Minimum duration is generally 5 days for community-acquired pneumonia and 7 days for hospital-acquired or ventilator-associated pneumonia; patients should be clinically stable with normal vital signs before therapy is discontinued (Ref).

Prosthetic joint infection

Prosthetic joint infection:

Note: For pathogen-directed therapy for methicillin-susceptible *S. aureus* (Ref)

IV: 2 g every 8 hours. Duration ranges from 2 to 6 weeks depending on prosthesis management, use of rifampin, and other patient-specific factors. Note: In select cases (eg, debridement and retention of prosthesis or one-stage arthroplasty), combine with oral rifampin and give oral suppressive antibiotic therapy following completion of IV treatment (Ref).

Septic arthritis

Septic arthritis:

Note: For pathogen-directed therapy for methicillin-susceptible *S. aureus* (Ref).

IV: 2 g every 8 hours. Total treatment duration is 3 to 4 weeks (in the absence of osteomyelitis, bacteremia, or other complicating conditions), including appropriate oral step-down therapy (Ref).

Skin and soft tissue infection

Skin and soft tissue infection:

Erysipelas or nonpurulent cellulitis in patients without risk for methicillin-resistant *S. aureus*: **IV:** 1 to 2 g every 8 hours. Total duration of therapy ≥ 5 days (including oral step-down therapy); may extend to 14 days depending on severity and clinical response (Ref).

Pathogen-directed therapy for methicillin-susceptible *S. aureus*: **IV:** 1 to 2 g every 8 hours. Total duration of therapy is 5 to 14 days (including oral step-down therapy) depending on severity of infection, need for debridement, and clinical response (Ref). **Note:** For necrotizing infections,

antibiotic therapy must be used in conjunction with early and aggressive surgical exploration and debridement of necrotic tissue; continue until further debridement is not necessary and the patient has clinically improved and is afebrile for 48 to 72 hours (Ref).

Surgical site incisional infection (trunk or extremity surgery, not involving axilla or perineum): IV: 1 g every 8 hours; duration is dependent upon severity, need for debridement, and clinical response (Ref).

Streptococcus, maternal prophylaxis for prevention of neonatal disease

Streptococcus (group B), maternal prophylaxis for prevention of neonatal disease (alternative agent) (off-label use):

IV: 2 g as a single dose at onset of labor or prelabor rupture of membranes, then 1 g every 8 hours until delivery. Reserve for patients with penicillin allergy who are at **low** risk for anaphylaxis (eg, rash without urticaria and no systemic symptoms) (Ref).

Surgical prophylaxis

Surgical prophylaxis: IV: 2 g for patients <120 kg or 3 g for patients ≥120 kg; administer within 60 minutes of surgical incision. Use in combination with metronidazole for procedures requiring anaerobic coverage (eg, colorectal and clean-contaminated head and neck procedures) (Ref). May use in combination with azithromycin for nonelective cesarean delivery during labor or after membrane rupture (Ref). May repeat dose intraoperatively in 4 hours if procedure is lengthy or if there is excessive blood loss (Ref); maximum dose: 12 g/day (Ref). In cases where an extension of prophylaxis is warranted postoperatively, total duration should be ≤24 hours (Ref). Postoperative prophylaxis is not recommended in clean and clean-contaminated surgeries (Ref).

Toxic shock syndrome

Toxic shock syndrome (off-label use):

Note: For pathogen-directed therapy for group A streptococcus (alternative agent for patients with nonsevere, non-IgE-mediated penicillin allergy) or methicillin-susceptible *S. aureus* (Ref).

IV: 2 g every 8 hours in combination with clindamycin. In the absence of bacteremia, treat for a total of ≥10 days, including oral step-down therapy (Ref).

Urinary tract infection, complicated

Urinary tract infection, complicated (pyelonephritis or urinary tract infection with systemic signs/symptoms): *Pathogen-directed therapy for susceptible organisms:* **IV:** 1 g every 8 hours (Ref). For patients with symptomatic improvement within the first ~48 hours of therapy, total duration (including oral step-down therapy if appropriate) is 5 to 7 days (7 days if therapy is completed with cefazolin) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: The following dose adjustments are for the usual recommended dosing of 1 to 2 g IV every 8 hours.

Altered kidney function (Ref): IV:

Note: Consider an initial unadjusted dose appropriate to the severity of the infection before reducing the dose.

CrCl ≥50 mL/minute: 1 to 2 g every 8 hours.

CrCl 30 to <50 mL/minute: 1 to 2 g every 8 to 12 hours.

CrCl >10 to <30 mL/minute: 500 mg to 1 g every 12 hours (some experts give 2 g every 12 hours for severe infections in patients with CrCl 10 to <30 mL/minute (Ref)).

CrCl ≤10 mL/minute: 500 mg to 1 g every 24 hours.

Augmented renal clearance (measured urinary CrCl ≥130 mL/minute/1.73 m²): Augmented renal clearance (ARC) is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations. Young patients (<55 years of age) admitted post trauma or major surgery are at highest risk for ARC, as well as those with sepsis, burns, or hematologic malignancies. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients (Ref).

IV: 2 g every 6 hours (Ref).

Hemodialysis, intermittent (thrice weekly): Dialyzable (45% to 60% (Ref)):

IV:

Daily dosing: 500 mg to 1 g every 24 hours (when scheduled dose falls on a dialysis day, administer after dialysis).

or

Thrice weekly (post dialysis) dosing: 2 g after dialysis 3 times weekly (Ref) **or** 20 mg/kg (maximum dose: 2 g) after dialysis 3 times weekly (Ref) **or** 2 g after dialysis if next dialysis is expected in 48 hours **or** 3 g after dialysis if next dialysis is expected in 72 hours (Ref).

Peritoneal dialysis: IV: 500 mg every 12 hours or 1 g every 24 hours (Ref).

CRRT: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection), organism MIC, residual kidney function, and consideration of initial loading doses. Close monitoring of response and adverse reactions (eg, neurotoxicity (Ref)) due to drug accumulation is important.

CVVH/CVVHD/CVVHDF: **IV:** 2 g loading dose followed by either 1 g every 8 hours or 2 g every 12 hours (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection), organism MIC, residual kidney function, and consideration of initial loading doses. Close monitoring of response and adverse reactions (eg, neurotoxicity (Ref)) due to drug accumulation is important.

IV: 2 g loading dose followed by either 1 g every 8 hours or 2 g every 12 hours (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Liver impairment prior to treatment initiation:

Child-Turcotte-Pugh class A to C: No dosage adjustment necessary (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, or 3 obesity (BMI ≥30 kg/m²) :

Surgical prophylaxis (>120 kg): **IV:** 3 g (Ref); may repeat dose intraoperatively in 4 hours if procedure is lengthy (eg, longer than 2 to 3 hours) or if there is excessive blood loss (Ref).

Indications other than surgical prophylaxis:

Traditional intermittent infusion method: Refer to indication-specific dosing; higher end of dosing ranges is recommended (Ref).

Continuous-infusion method: **IV:** 2 g loading dose (Ref), then 6 to 8 g over 24 hours (Ref).

Rationale for recommendations:

There are limited data evaluating the effect of obesity on dosing requirements for cefazolin outside of perioperative use. The available perioperative data to support higher doses for patients weighing >120 kg are based on pharmacokinetic modeling rather than clinical outcomes (Ref). Although some data demonstrate 2 g may be sufficient for perioperative use in patients with obesity, current guidelines recommend 3 g for patients >120 kg in the perioperative setting (Ref). Use of 3 g for indications other than surgical prophylaxis is not currently recommended.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Cefazolin: Pediatric drug information")

General dosing:

Infants, Children, and Adolescents: IM, IV: 25 to 100 mg/kg/day divided every 8 hours; usual maximum dose: 2,000 mg/dose (Ref). For serious infections, may use up to 150 mg/kg/day divided every 6 to 8 hours, not to exceed 12 **g/day** (Ref). Continuous infusion administration may be recommended for infections due to isolates with elevated MICs or in patients with critical illness or augmented renal clearance (Ref).

Endocarditis, prophylaxis before invasive dental procedures

Endocarditis, prophylaxis before invasive dental procedures (alternative agent): Limited data available:

Note: Alternative agent for use in patients who are unable to take oral medication and who have penicillin or ampicillin allergy (excluding those with a history of anaphylaxis, angioedema, or urticaria) (Ref). Recommended only in patients who are at highest risk for infective endocarditis (IE) or adverse outcomes (eg, history of IE, cardiac valve repair using prosthetic valves or material, unrepaired cyanotic congenital heart disease [CHD], left ventricular assist device or implantable heart, repaired CHD with prosthetic material or device during first 6 months after procedure, pulmonary artery valve or conduit placement [eg, Melody valve, Contegra conduit], repaired CHD with residual defects at the site or adjacent to site of prosthetic patch or device, heart transplant recipients with cardiac valvulopathy) (Ref).

Infants, Children, and Adolescents: IM, IV: 50 mg/kg as a single dose administered 30 to 60 minutes prior to dental procedure; maximum dose: 1,000 mg/dose (Ref).

Endocarditis, treatment

Endocarditis, treatment: Children and Adolescents: IV: 100 mg/kg/day in divided doses every 8 hours; usual adult dose: 2,000 mg/dose; maximum daily dose: 12 **g/day**; treat for at least 4 weeks; longer durations may be necessary; may use with or without gentamicin (Ref).

Osteoarticular infection, acute

Osteoarticular infection, acute (eg, bacterial arthritis, osteomyelitis): Infants, Children, and Adolescents: IV: 100 to 150 mg/kg/day divided every 6 to 8 hours; usual adult dose: 2,000 mg/dose; maximum daily dose: 12 **g/day** (Ref). Minimum total duration is 10 to 14 days for bacterial arthritis and 21 to 28 days for osteomyelitis; however, duration may be longer and should be individualized based on several factors, including causative pathogen, response to therapy, and normalization of inflammatory markers (Ref).

Peritonitis

Peritonitis (peritoneal dialysis): Limited data available: Infants, Children, and Adolescents:

Prophylaxis (Ref):

Touch contamination of PD line: Intraperitoneal: 125 mg per liter.

Invasive dental procedures: IV: 25 mg/kg administered 30 to 60 minutes before procedure; maximum dose: 1,000 mg/dose.

Gastrointestinal or genitourinary procedures: IV: 25 mg/kg administered 60 minutes before procedure; maximum dose: 2,000 mg/dose.

Treatment (Ref):

Intermittent: Intraperitoneal: 20 mg/kg every 24 hours in the long dwell.

Continuous: Intraperitoneal: Loading dose: 500 mg per liter of dialysate; maintenance: 125 mg per liter of dialysate.

Pneumonia, community acquired

Pneumonia, community acquired:

Note: Not recommended for empiric treatment; recommended as pathogen-directed therapy for methicillin-susceptible *Staphylococcus aureus* (MSSA) (Ref).

Infants >3 months, Children, and Adolescents: IV: 150 mg/kg/day divided every 8 hours (Ref); usual maximum dose: 2,000 mg/dose (Ref). Guidelines indicate that a duration of 10 days of therapy is best studied (Ref). More recent data support a shorter duration of 5 days in patients with uncomplicated disease who respond quickly to therapy; however, studies evaluated empiric pneumonia treatment, not treatment specific to MSSA or cefazolin (eg, first-line therapy was amoxicillin for empiric treatment) (Ref).

Skin and soft tissue infection

Skin and soft tissue infection:

Note: Not appropriate as monotherapy if methicillin-resistant *S. aureus* (MRSA) is suspected or confirmed.

Infants, Children, and Adolescents: IV: 50 to 100 mg/kg/day divided every 8 hours; doses on the higher end of the range recommended for severe infections (eg, necrotizing fasciitis caused by methicillin-susceptible *S. aureus*); maximum dose: 2,000 mg/dose. Duration should be individualized based on resolution of signs and symptoms; typical total duration is 5 to 7 days, but may be longer, particularly in severe cases or pyomyositis (eg, cellulitis: may need up to 10 days; pyomyositis: 14 to 21 days) (Ref).

Surgical prophylaxis

Surgical prophylaxis: Infants, Children, and Adolescents: IV: 30 mg/kg within 60 minutes prior to procedure, may repeat in 4 hours for prolonged procedure or excessive blood loss (eg, >1,500 mL in adults); maximum dose dependent upon patient weight: Weight <120 kg: 2,000 mg/dose; weight ≥120 kg: 3,000 mg/dose (Ref).

Urinary tract infection

Urinary tract infection (UTI): Infants, Children, and Adolescents: IV: 50 mg/kg/day in divided doses every 8 hours (Ref). Usual adult dose: 1,000 mg/dose (Ref). Duration of therapy should be individualized based on patient age, severity/extent of infection, and clinical response; for uncomplicated cystitis in patients ≥3 months of age, 5 days of treatment may be adequate, though patients <2 years of age may require a longer course (eg, 7 days) (Ref). For complicated UTI, including pyelonephritis, treat for a total of 6 to 10 days; while 7 to 14 days has been recommended for complicated UTI, this was not shown to improve outcomes compared to a shorter duration of 6 to 10 days (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Note: Dosing is based on pharmacokinetic parameters, limited pediatric studies, adult studies, and expert opinion (Ref).

The following dose adjustments assume a usual recommended dose of 25 to 50 mg/kg/dose every 8 hours.

Altered kidney function:

Infants, Children, and Adolescents:

GFR $\geq$ 50 mL/minute/1.73 m²: IV, IM: No dosage adjustment necessary.

GFR 30 to <50 mL/minute/1.73 m²: IV, IM: 25 to 30 mg/kg/dose every 12 hours; doses up to 50 mg/kg/dose may be needed for severe infections; maximum dose: 2,000 mg/dose.

GFR 10 to 30 mL/minute/1.73 m²: IV, IM: 25 to 30 mg/kg/dose every 24 hours; doses up to 50 mg/kg/dose may be needed for severe infections; maximum dose: 2,000 mg/dose.

GFR $\leq$ 10 mL/minute/1.73 m²: IV, IM: 25 to 30 mg/kg/dose every 48 hours; doses up to 50 mg/kg/dose may be needed for severe infections; maximum dose: 2,000 mg/dose.

Hemodialysis, intermittent: Dialyzable: 35% to 65%.

Note: Appropriate dosing requires consideration of drug penetration to site of infection, minimum inhibitory concentration (MIC) of bacteria, severity of illness, residual kidney function, and interval between dialysis sessions.

Infants, Children, and Adolescents:

Intermittent dosing (eg, 3 times weekly): IV: 25 to 50 mg/kg/dose after dialysis; maximum dose: 2,000 mg/dose. **Note:** Children with residual kidney function may require higher or more frequent dosing.

Peritoneal dialysis:

Infants, Children, Adolescents: IV, IM: 25 to 30 mg/kg/dose every 24 to 48 hours; maximum dose: 1,000 mg/dose.

CRRT:

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of $\geq$ 1,500 mL/m²/hour unless otherwise noted; flow rates vary widely in pediatric patients. Appropriate dosing requires consideration of drug penetration to site of infection, MIC of bacteria, and severity of illness. Close monitoring of response and adverse reactions due to drug accumulation (eg, neurotoxicity) is important. Due to minimal data in pediatric patients receiving CRRT, consider monitoring serum concentrations (eg, trough concentration) if available. Dosing based on limited adult information and expert opinion.

CVVH/CVVHD/CVVHDF: Infants, Children, and Adolescents: IV: 25 to 50 mg/kg/dose every 8 to 12 hours. Maximum dose: 2,000 mg/dose.

Augmented renal clearance (ARC):

Note: ARC is a condition that occurs in certain critically ill patients without organ dysfunction and with normal serum creatinine concentrations that results in increased drug elimination. An 8- to 24-hour measured urinary CrCl is necessary to identify these patients.

Infants, Children, and Adolescents:

GFR $\geq$ 200 mL/minute/1.73 m²:

Continuous infusion: 150 mg/kg/day as a continuous infusion; maximum daily dose: 12 g/day. May give an initial loading dose of 30 mg/kg (maximum dose: 2,000 mg/dose) before starting the continuous infusion if rapid attainment of therapeutic drug concentrations is desired (eg, sepsis). Dosing is

based on a pharmacokinetic modeling study; additional dosages (eg, lower daily doses, every-6-hour dosing) may be appropriate depending on the clinical situation.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions (Significant): Considerations

Clostridioides difficile infection

Clostridioides difficile infection has occurred, including ***Clostridioides difficile*-associated diarrhea** and ***Clostridioides difficile* colitis** (Ref).

Mechanism: Dose- and time-related; potentially related to recent or cumulative antibiotic exposure. Cefazolin may cause disruption of the intestinal microbiota resulting in the overgrowth of pathogens, such as *C. difficile* (Ref).

Onset: Varied; may start on the first day of antibiotic therapy or up to 3 months postantibiotic (Ref).

Risk factors:

- Exposure to cefazolin during the 4-week period prior to positive *C. difficile* test (Ref)
- Antibiotic exposure (highest risk factor) (Ref)
- Type of antibiotic (third-/fourth-generation cephalosporins among the highest risk (Ref)
- Long durations in a hospitalization or other healthcare setting (recent or current) (Ref)
- Older adults (Ref)
- Immunocompromised conditions (Ref)
- A serious underlying condition (Ref)
- GI surgery/manipulation (Ref)
- Antiulcer medications (eg, proton pump inhibitors, H₂ blockers) (Ref)
- Chemotherapy (Ref)

Hemolytic anemia

Hemolytic anemia has occurred in patients receiving cephalosporins, including cefazolin (Ref). While extremely rare, hemolytic anemia associated with cefazolin has been fatal (Ref).

Mechanism: Not clearly established; no clear consensus about the mechanism of cefazolin-induced hemolytic anemia among limited case reports (Ref).

Onset: Varied; occurs within the first few hours to 7 days after the first dose (Ref).

Hypersensitivity reactions (immediate and delayed)

Hypersensitivity reactions (immediate and delayed) range from morbilliform **skin rash** to rare cases of **anaphylaxis** and **anaphylactic shock** (Ref). **Urticaria** and **angioedema** have also occurred (Ref). Delayed hypersensitivity reactions, including severe cutaneous adverse reactions (SCARs) (**acute generalized exanthematous pustulosis** [AGEP], **Stevens-Johnson syndrome** [SJS], and **toxic epidermal necrolysis** [TEN]) and **serum sickness-like reaction**, have been reported (Ref). The incidence of cefazolin hypersensitivity in perioperative settings is low (<1%) (Ref). Due to the unique R1-side chain, the incidence of cefazolin hypersensitivity is lower than other beta-lactam antibiotics. For this reason, beta-lactam cross-reactivity is rare, and patients may experience "selective hypersensitivity" to cefazolin (Ref).

Mechanism: Non-dose-related; immunologic. Immediate hypersensitivity reactions (eg, anaphylaxis, urticaria) are IgE-mediated. Delayed hypersensitivity reactions, including maculopapular rash and SCARs, are T-cell-mediated (Ref).

Onset: Immediate hypersensitivity reactions: Rapid; occur within 1 hour of administration but may occur up to 6 hours after exposure (Ref). Delayed hypersensitivity reactions: Maculopapular reactions: Intermediate; occur 7 to 10 days after initiation. Other reactions (including SCARs): Varied; occur after 7 to 14 days up to 3 months (Ref).

Risk factors:

- Cross-reactivity between penicillins and cephalosporins and among cephalosporins is mostly related to side chain similarity (Ref). Studies and a practice parameter have shown negligible cross-reactivity between penicillins and cefazolin (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Hypotension, syncope

Dermatologic: Pruritus (including genital pruritus), skin rash

Gastrointestinal: Abdominal cramps, anorexia, diarrhea, epigastric pain, flatulence, heartburn, nausea, oral candidiasis, oral mucosal ulcer, pruritus ani, vomiting

Genitourinary: Vaginitis, vulvovaginal candidiasis, vulvovaginal pruritus

Hematologic & oncologic: Eosinophilia, leukopenia, neutropenia, thrombocytopenia, thrombocytosis

Hepatic: Hepatitis, increased serum alanine aminotransferase, increased serum alkaline phosphatase, increased serum aspartate aminotransferase

Local: Induration at injection site, injection-site phlebitis, pain at injection site

Nervous system: Asthenia, confusion, dizziness, drowsiness, drug fever, fatigue, headache

Renal: Increased blood urea nitrogen, increased serum creatinine, renal failure syndrome

Postmarketing:

Dermatologic: Acute generalized exanthematous pustulosis (Cleminson 2020), Stevens-Johnson syndrome (Saavedra 2012), toxic epidermal necrolysis (Sriganesh 2018), urticaria (Pipet 2011)

Gastrointestinal: *Clostridioides difficile*-associated diarrhea (Hansen 2013), *Clostridioides difficile* colitis (Hansen 2013)

Hematologic & oncologic: Hemolytic anemia (Mause 2021)

Hypersensitivity: Anaphylactic shock (Bogas 2021), anaphylaxis (Bogas 2021), angioedema (Pipet 2011), serum sickness-like reaction (Nguyen 2017)

Renal: Acute interstitial nephritis

Contraindications

Immediate hypersensitivity (eg, anaphylaxis, serious skin reactions) to cefazolin, other cephalosporin antibiotics, penicillins, other beta-lactams, or any component of the formulation.

Warnings/Precautions

Concerns related to adverse effects:

- Elevated INR: May be associated with increased INR, especially in nutritionally deficient patients, prolonged treatment, hepatic or renal disease.
- Superinfection: Prolonged use may result in fungal or bacterial superinfection.

Disease-related concerns:

- Gastrointestinal disease: Use with caution in patients with a history of gastrointestinal disease, particularly colitis.
- Renal impairment: Use with caution in patients with renal impairment; dosage adjustment required.
- Seizure disorders: Use with caution in patients with a history of seizure disorder; high levels, particularly in the presence of renal impairment, may increase risk of seizures.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous [preservative free]:

Generic: 1 g/50 mL in Dextrose 4% (50 mL); 2 g/100 mL in Dextrose 4% (100 mL); 3 g/150 mL in Dextrose 4% (150 mL)

Solution Reconstituted, Injection:

Generic: 1 g (1 ea); 10 g (1 ea)

Solution Reconstituted, Injection [preservative free]:

Generic: 500 mg (1 ea); 1 g (1 ea); 10 g (1 ea); 100 g (1 ea); 300 g (1 ea)

Solution Reconstituted, Intravenous [preservative free]:

Generic: 1 g (1 ea [DSC]); 1 g and Dextrose 4% (1 ea); 2 g and Dextrose 3% (1 ea); 3 g and Dextrose 2% (1 ea)

Solution Reconstituted, Injection, as sodium [preservative free]:

Generic: 2 g (1 ea); 3 g (1 ea)

Solution Reconstituted, Intravenous, as sodium:

Generic: 2 g (1 ea); 3 g (1 ea)

Solution Reconstituted, Intravenous, as sodium [preservative free]:

Generic: 2 g (1 ea); 3 g (1 ea)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (ceFAZolin Sodium-Dextrose Intravenous)

1GM/50ML 4% (per mL): \$0.13

2GM/100ML 4% (per mL): \$0.11

3GM/150ML 4% (per mL): \$0.11

Solution (reconstituted) (ceFAZolin Sodium Injection)

1 g (per each): \$1.20 - \$2.61

2 g (per each): \$7.31

3 g (per each): \$10.97

10 g (per each): \$8.11 - \$60.00

100 g (per each): \$144.00

300 g (per each): \$420.00

500 mg (per each): \$1.92 - \$3.11

Solution (reconstituted) (ceFAZolin Sodium Intravenous)

2 g (per each): \$5.86 - \$7.02

3 g (per each): \$7.08 - \$10.53

Solution (reconstituted) (ceFAZolin Sodium-Dextrose Intravenous)

1GM 4%(50ML) (per each): \$16.65

2GM 3%(50ML) (per each): \$18.73

3GM 2%(50ML) (per each): \$18.12

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution Reconstituted, Injection:

Generic: 500 mg (1 ea); 1 g (1 ea); 10 g (1 ea); 100 g (1 ea)

Solution Reconstituted, Intravenous:

Generic: 10 g (1 ea)

Solution Reconstituted, Injection, as sodium:

Generic: 2 g (1 ea)

Administration: Adult

Note: The approved route of administration (eg, IV push) may vary based on product and/or vial size; refer to product-specific labeling and/or institution-specific policies.

IM: Inject deep IM into large muscle mass.

IV: Inject direct IV over 3 to 5 minutes, as an intermittent infusion over 30 to 60 minutes, or as a continuous infusion. **Note:** The manufacturer's labeling from Hikma Pharmaceuticals and WG Critical Care states their 2 g vial should be given over 7 to 11 minutes if administering as a direct IV injection.

Some penicillins (eg, carbenicillin, ticarcillin and piperacillin) have been shown to inactivate aminoglycosides *in vitro*. This has been observed to a greater extent with tobramycin and gentamicin, while amikacin has shown greater stability against inactivation. Concurrent use of these agents may pose a risk of reduced antibacterial efficacy *in vivo*, particularly in the setting of profound renal impairment. However, definitive clinical evidence is lacking. If combination penicillin/aminoglycoside therapy is desired in a patient with renal dysfunction, separation of doses (if feasible), and routine monitoring of aminoglycoside levels, CBC, and clinical response should be considered.

Intraperitoneal (off-label route): May administer intermittently or continuously (with every exchange) (Ref).

Preparation for Administration: Adult

Note: Preparation instructions may vary based on vial size, administration method, and/or manufacturer labeling; refer to product-specific labeling and/or institution-specific policies.

IM: Reconstitute 500 mg vial with 2 mL SWFI, 1 g vial with 2.5 mL SWFI, 2 g vial with 5 mL SWFI, and 3 g vial with 7.5 mL SWFI, resulting in a concentration of 225 mg/mL, 330 mg/mL, 330 mg/mL, and 330 mg/mL, respectively. Shake well.

IV Push: Reconstitute 500 mg vial with 2 mL SWFI, 1 g vial with 2.5 mL SWFI, 2 g vial with 5 or 5.5 mL SWFI (varies by manufacturer), and 3 g vial with 7.5 mL SWFI; shake well. Reconstituted solution may be directly injected after further dilution with 5 or 11 mL SWFI (varies by manufacturer).

Intermittent or continuous IV infusion: Reconstitute 500 mg vial with 2 mL SWFI, 1 g vial with 2.5 mL SWFI, 2 g vial with 5 or 15 mL SWFI (varies by manufacturer), 3 g vial with 7.5 or 15 mL SWFI (varies by manufacturer), and 10 g vial with 45 mL of SWFI, bacteriostatic water for injection, or NS to yield 1 g/5 mL or with 96 mL of SWFI, bacteriostatic water for injection, or NS to yield 1 g/10 mL. Further dilute dose in 50 to 100 mL of D5W or NS. Other compatible solutions may be used; refer to product-specific labeling.

Administration: Pediatric

Parenteral:

IM: Deep IM injection into a large muscle mass.

IV: Infuse over 30 to 60 minutes; may also be administered IV push over 3 to 5 minutes (Ref) or as a continuous infusion. **Note:** The manufacturer's labeling from Hikma Pharmaceuticals states their 2 g dose should be given over 7 to 11 minutes if administering as a direct IV injection.

Preparation for Administration: Pediatric

Parenteral:

IM:

Reconstitute vial with appropriate amount of SWFI based on the following table. Shake well.

Cefazolin Single-Dose Vial Reconstitution Volumes for IM Administration^a

Vial size

SWFI diluent volume

Approximate withdrawable volume

Final concentration

^a Suggested diluent volumes and final concentrations may vary based on manufacturer; refer to product labeling.

500 mg

2 mL

2.2 mL

225 mg/mL

1,000 mg

2.5 mL

3 mL

330 mg/mL

2,000 mg

5 mL

6 mL

330 mg/mL

3,000 mg

7.5 mL

9 mL

330 mg/mL

IV push:

Reconstitute vial with appropriate amount of SWFI based on the following table. Withdraw dose from vial and further dilute with 5 mL SWFI prior to administration. **Note:** Dilution of dose in SWFI to a concentration of 138 mg/mL results in an osmolality that is suitable for peripheral infusion in fluid restricted patients (Ref).

Cefazolin Single-Dose Vial Reconstitution Volumes for IV Push Administration (Further Dilution Required)^a

Vial size

SWFI diluent volume

Resultant vial concentration

^a Suggested diluent volumes and final concentrations may vary based on manufacturer; refer to product labeling.

500 mg

2 mL

225 mg/mL

1,000 mg

2.5 mL

330 mg/mL

2,000 mg

5 mL

330 mg/mL

3,000 mg

7.5 mL

330 mg/mL

Intermittent or continuous IV infusion:

Single-dose vials: Reconstitute vial with appropriate amount of SWFI based on the following table. Withdraw dose from vial and further dilute dose in a compatible solution (eg, D5W, NS) to a final concentration of 5 to 40 mg/mL per the manufacturer; concentrations as high as 100 mg/mL have been reported (Ref). For neonatal patients, Vermont Oxford Network recommends a concentration of 100 mg/mL (Ref).

Cefazolin Single-Dose Vial Reconstitution Volumes for IV Infusion (Further Dilution Required)^a

Vial size

SWFI diluent volume

Resultant vial concentration

^a Suggested diluent volumes and final concentrations may vary based on manufacturer; refer to product labeling for specific information.

500 mg

2 mL

225 mg/mL

1,000 mg

2.5 mL

330 mg/mL

2,000 mg

5 mL

317.5 to 330 mg/mL

15 mL

136 mg/mL

3,000 mg

7.5 mL

330 mg/mL

15 mL

196 mg/mL

Bulk vials: Pharmacy bulk vial (10 g) may be diluted with 45 mL or 96 mL of SWFI or NS, resulting in concentrations of ~200 mg/mL or 100 mg/mL, respectively. Withdraw dose from vial and further dilute in a compatible solution (eg, D5W, NS) to a final concentration of 5 to 40 mg/mL per the manufacturer; concentrations as high as 100 mg/mL have been reported (Ref).

Storage/Stability

Store intact vials at 20°C to 25°C (68°F to 77°F). Reconstituted solutions of cefazolin are light yellow to yellow. Protection from light is recommended for the powder and for the reconstituted solutions. Reconstituted vials and solutions diluted for infusion are stable for 4 to 24 hours at room temperature and for 3 to 10 days under refrigeration (varies by manufacturer; refer to product-specific labeling).

DUPLEX: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F) prior to activation; do not freeze. Following activation, stable for 24 hours at room temperature and for 7 days under refrigeration.

GALAXY: Store at or below -20°C (-4°F). Thawed solution stable for 48 hours at room temperature and for 30 days under refrigeration. Do not refreeze.

Use: Labeled Indications

Biliary tract infection: Treatment of biliary tract infections due to *Escherichia coli*, various strains of streptococci, *Proteus mirabilis*, *Klebsiella* species, and *Staphylococcus aureus*.

Bloodstream infection: Treatment of bloodstream infection due to methicillin-susceptible staphylococci, *E. coli*, *P. mirabilis*, and *Klebsiella* species.

Bone and joint infection: Treatment of bone and joint infections due to *S. aureus*.

Endocarditis, treatment: Treatment of endocarditis due to methicillin-susceptible staphylococci and group A beta-hemolytic streptococci (*Streptococcus pyogenes*).

Genital infection: Treatment of genital infections (ie, prostatitis, epididymitis) due to *E. coli*, *P. mirabilis*, and *Klebsiella* species.

Respiratory tract infection: Treatment of respiratory tract infections due to *Streptococcus pneumoniae*, *Klebsiella* species, methicillin-susceptible *S. aureus*, and group A beta-hemolytic streptococci.

Skin and soft tissue infection: Treatment of skin and soft tissue infections due to methicillin-susceptible *S. aureus*, group A beta-hemolytic streptococci, and other strains of streptococci.

Surgical prophylaxis: To reduce the incidence of certain postoperative infections in adults and pediatric patients 10 to 17 years of age undergoing surgical procedures.

Urinary tract infection: Treatment of urinary tract infections due to *E. coli*, *P. mirabilis*, and *Klebsiella* species.

Use: Off-Label: Adult

Endocarditis, prophylaxis; Peritonitis, treatment, peritoneal dialysis; Streptococcus (group B), maternal prophylaxis for prevention of neonatal disease; Toxic shock syndrome

Medication Safety Issues

Sound-alike/look-alike issues:

CeFAZolin may be confused with cefoTEtan, cefOXitin, cefprozil, cefTAZidime, cefTRIAxone, cephalexin.

Older Adult: High-Risk Medication:

Cefazolin is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) for use in asymptomatic bacteriuria (O'Mahony 2023).

Metabolism/Transport Effects

Substrate of OAT1/3

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Aescin: Antibiotics may increase serum concentrations of Aescin. Management: Consider alternatives to this combination as aescin product labeling states coadministration with antibiotics is not recommended. *Risk D: Consider Therapy Modification*

Aminoglycosides: Cephalosporins may increase nephrotoxic effects of Aminoglycosides. Cephalosporins may decrease serum concentrations of Aminoglycosides. *Risk C: Monitor*

Bacillus Calmette Guerin (BCG) (Intravesical): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Intravesical). *Risk X: Avoid*

Bacillus Calmette Guerin (BCG) (Percutaneous): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Percutaneous). *Risk C: Monitor*

Bacillus clausii: Antibiotics may decrease therapeutic effects of Bacillus clausii. Management: Bacillus clausii should be taken in between antibiotic doses during concomitant therapy. *Risk D: Consider Therapy Modification*

Cholera Vaccine: Antibiotics may decrease therapeutic effects of Cholera Vaccine. Management: Avoid cholera vaccine in patients receiving systemic antibiotics, and within 14 days following the use of oral or parenteral antibiotics. *Risk X: Avoid*

Fecal Microbiota (Live) (Oral): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Oral). *Risk X: Avoid*

Fecal Microbiota (Live) (Rectal): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Rectal). *Risk X: Avoid*

Fosphenytoin: CeFAZolin may increase serum concentrations of Fosphenytoin. Specifically, the ratio of free phenytoin to total phenytoin may be increased. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Antibiotics may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Lactobacillus and Estriol: Antibiotics may decrease therapeutic effects of Lactobacillus and Estriol. *Risk C: Monitor*

Mycophenolate: Antibiotics may decrease active metabolite exposure of Mycophenolate. Specifically, concentrations of mycophenolic acid (MPA) may be reduced. *Risk C: Monitor*

Phenytoin: CeFAZolin may increase serum concentrations of Phenytoin. Specifically, the ratio of free phenytoin to total phenytoin may be increased. *Risk C: Monitor*

Probenecid: May increase serum concentrations of Cephalosporins. *Risk C: Monitor*

RifAMPin: CeFAZolin may increase adverse/toxic effects of RifAMPin. Specifically, the risk for bleeding may be increased. Management: Avoid concomitant use of rifampin with cefazolin when possible. If combined, closely monitor prothrombin time or other coagulation tests and administer vitamin K as needed. *Risk D: Consider Therapy Modification*

Sodium Picosulfate: Antibiotics may decrease therapeutic effects of Sodium Picosulfate. Management: Consider using an alternative product for bowel cleansing prior to a colonoscopy in patients who have recently used or are concurrently using an antibiotic. *Risk D: Consider Therapy Modification*

Typhoid Vaccine: Antibiotics may decrease therapeutic effects of Typhoid Vaccine. Only the live attenuated Ty21a strain is affected. Management: Avoid use of live attenuated typhoid vaccine (Ty21a) in patients being treated with systemic antibacterial agents. Postpone vaccination until 3 days after cessation of antibiotics and avoid starting antibiotics within 3 days of last vaccine dose. *Risk D: Consider Therapy Modification*

Vitamin K Antagonists: Cephalosporins may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Pregnancy Considerations

Cefazolin crosses the placenta.

An increased risk of major birth defects or other adverse fetal or maternal outcomes has generally not been observed following use of cephalosporins, including cefazolin, during pregnancy. Based on available data, cephalosporin antibiotics are generally considered compatible for use during pregnancy (Ailes 2016; Bookstaver 2015; Crider 2009; Czeizel 2001; Lamont 2014; Muanda 2017a; Muanda 2017b).

Due to pregnancy-induced physiologic changes, some pharmacokinetic parameters of cefazolin may be altered (Allegaert 2009; Allegaert 2020; Elkayal 2021; Elkomy 2014; Fouks 2022; Philipson 1987). In addition to pregnancy, obesity has been found to influence the pharmacokinetics of cefazolin (Ahmadzia 2015; Eley 2020; Groff 2017; Kram 2017; Pevzner 2011; Stitely 2013; Young 2015). Higher doses are recommended in obese patients; however, additional studies are needed to determine the ideal dose, especially in pregnant patients with extreme obesity (ACOG 2018; Kominiarek 2024; La Rosa 2020; Mackeen 2024).

Cefazolin is recommended prior to all cesarean deliveries unless the patient is already receiving an appropriate antibiotic. A single dose administered within 60 minutes prior to skin incision is recommended unless drug allergies are present. Dose adjustments of cefazolin may be considered based on maternal weight (ACOG 2018; Mackeen 2024).

Cefazolin is recommended as an alternative antibiotic for group B streptococcus (GBS) prophylaxis in pregnant patients who are penicillin allergic and at low risk for anaphylaxis. Treatment is intended

to prevent early-onset disease in newborns and is most effective when initiated at least 4 hours prior to delivery. Prophylaxis is reserved for pregnant patients with a positive GBS vaginal or rectal screening in late gestation; GBS bacteriuria during the current pregnancy; history of birth of an infant with early-onset GBS disease; and unknown GBS culture status with any of the following: birth <37 0/7 weeks gestation, intrapartum fever, prolonged rupture of membranes, known GBS positive in a previous pregnancy, or intrapartum nucleic acid amplification testing positive for GBS (ACOG 2020).

Cefazolin is one of the recommended antibiotics that may be used prior to vaginal delivery in patients at high risk for endocarditis. This includes patients with congenital heart disease, prosthetic valves, previous infective endocarditis, or cardiac transplant. Cefazolin is given 30 to 60 minutes prior to a vaginal delivery; the endocarditis prophylactic dose is the same as nonpregnant patients (ACOG 2018).

Breastfeeding Considerations

Cefazolin is present in breast milk.

- A study reports limited transfer of cefazolin into breast milk following administration of cefazolin 500 mg IM to 4 patients. Breast milk was sampled every hour for 12 hours after administration. Only trace amounts of cefazolin were present (limit of detection not stated). Cefazolin was not present in the serum of the breastfed infants. Although GA was not noted, pregnant patients who required an antibiotic for obstetrical issues near delivery were included in the study (Fukada 1973).

- Breast milk was sampled in 20 lactating mothers receiving a dose of cefazolin 2 g IV for postpartum and other infections. Two hours after the dose, the average maternal serum concentration of cefazolin was 54.33 mcg/mL. Average breast milk concentrations were 1.25, 1.51, and 1.16 mcg/mL at 2, 3, and 4 hours after the dose. The authors estimated that 1.5 mg/day of cefazolin would appear in the mother's milk, corresponding to <0.075% of the maternal dose, based on 1 L of milk production per day (Yoshioka 1979). Using a breast milk concentration of 1.51 mcg/mL, the estimated exposure of cefazolin to the breastfeeding infant would be 0.2265 mcg/kg/day (relative infant dose <1% based on the weight-adjusted maternal dose of 2 g/day).

- In general, breastfeeding is considered acceptable when the relative infant dose of a medication is <10% (Anderson 2016; Ito 2000).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. In general, antibiotics that are present in breast milk may cause non-dose-related modification of bowel flora; monitor the infant for GI disturbances (WHO 2002).

Dietary Considerations

Some products may contain sodium.

Monitoring Parameters

Renal function, hepatic function, CBC, signs of anaphylaxis during first dose.

Mechanism of Action

Inhibits bacterial cell wall synthesis by binding to one or more of the penicillin-binding proteins (PBPs) which in turn inhibits the final transpeptidation step of peptidoglycan synthesis in bacterial cell walls, thus inhibiting cell wall biosynthesis. Bacteria eventually lyse due to ongoing activity of cell wall autolytic enzymes (autolysins and murein hydrolases) while cell wall assembly is arrested.

Pharmacokinetics (Adult Data Unless Noted)

Distribution: Widely into most body tissues and fluids including gallbladder, liver, kidneys, bone, sputum, bile, pleural, and synovial; CSF penetration is low.

V_d :

Preterm neonates (GA: 25 to <32 weeks; PNA: Median: 16 days [range: 3 to 91 days]): 0.39 L/kg (range: 0.31 to 0.52 L/kg) (Balevic 2019).

Preterm and term neonates (GA: 30 to 40 weeks; PNA: 2 to 28 days): 0.212 to 0.373 L/kg (Deguchi 1988).

Children 3 to 12 years of age: 0.133 ± 0.015 L/kg (Koshida 1987).

Adults: 0.193 ± 0.064 L/kg (Scheld 1981).

Bone:serum concentration ratio: Median: 0.25 (range: 0.06 to 0.41) (Zeller 2009).

Protein binding:

Neonates (GA: 30 to 40 weeks; PNA: 2 to 28 days): 49% (range: 17% to 78%) (Deguchi 1988); protein binding is decreased (with a resulting higher free fraction) with lower albumin concentrations, higher bilirubin concentration, higher total cefazolin concentration, and lower PMA (Smits 2012).

Children 3 to 12 years: $78\% \pm 2.5\%$ (range: 75% to 82.3%) (Koshida 1987).

Adults: 80% (Marshall 1999).

Half-life elimination:

Preterm neonates (PNA: 2 to 12 days): IV: 4.05 ± 0.72 hours (range: 2.8 to 4.74 hours) (Sakata 1980).

Infants ≥ 9 months of age and children ≤ 10 years of age: IV: 1.7 ± 0.6 hours (Nahata 1991).

Children 10 to 12 years of age: IV: 1.95 hours (range 1.44 to 2.88 hours) (Schmitz 2015).

Adults: IV: 1.8 hours; IM: ~ 2 hours (prolonged with renal impairment).

Time to peak, serum: IM: 0.5 to 2 hours; IV: Within 5 minutes.

Excretion: Urine (70% to 80% as unchanged drug).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Anti-infective considerations:

Parameters associated with efficacy: Time dependent, associated with time free drug concentration (fT) > minimum inhibitory concentration (MIC).

Organism specific:

S. aureus (methicillin-susceptible [MSSA]): Goal: $\geq \sim 24\%$ to 55% $fT > MIC$ (bacteriostatic) (Turnidge 1998; Zelenitsky 2018); $\geq \sim 55\%$ to 100% $fT > MIC$ (bacteriocidal) (Vogelman 1988; Zelenitsky 2018).

E. coli: Goal: $\geq 60\%$ $fT > MIC$ (bacteriostatic); $\geq 100\%$ $fT > MIC$ (2-log kill) (Vogelman 1988).

Population specific:

Critically ill patients in the ICU: Minimum goal: $\geq 50\%$ $fT > MIC$; preferred goal: $\geq 100\%$ $fT > MIC$ (Roberts 2015); some experts favor $\geq 100\%$ $fT \geq 4$ times the MIC (Guilhaumou 2019).

Expected drug concentrations in patients with normal renal function:

Infants ≥ 9 months of age and children ≤ 10 years of age: C_{max} (peak), single dose: Mean dose: 22.7 mg/kg ± 4 mg/kg: 137 ± 87.9 mg/L (Nahata 1991).

Adults: C_{max} (peak), steady state: 1 g every 8 hours: 94 ± 30.33 mg/L (Bhalodi 2013).

Postantibiotic effect: Generally little to no postantibiotic effect (< 1 hour) for streptococci and gram-negative bacilli; for *S. aureus*: ~ 4 hours (Craig 1993; Craig 1998).

Patient Counseling Points

(For additional information see "Cefazolin: Patient drug information")

What is this drug used for?

- It is used to treat or prevent bacterial infections.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Diarrhea, upset stomach, or throwing up
- Decreased appetite
- Stomach pain or heartburn
- Gas
- Headache
- Feeling sleepy

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- Stevens-Johnson syndrome/toxic epidermal necrolysis like red, swollen, blistered, or peeling skin (with or without fever); red or irritated eyes; or sores in the mouth, throat, nose, or eyes
- Redness or white patches in mouth or throat
- Fever, chills, or sore throat; any unexplained bruising or bleeding; or feeling very tired or weak
- Seizures
- Anal irritation
- Vaginal itching or discharge
- Dizziness or passing out
- Feeling confused
- Redness or swelling where the injection is given
- C diff-associated diarrhea (CDAD) like stomach pain, cramps, or very loose, watery, or bloody stools
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Cefamezin | Kefzol | Totacef | Zepilen;
- (AR) Argentina: Cefalomicina | Cefazolina | Cefazolina argentina | Cefazolina Drawer | Cefazolina fabra | Cefazolina kilab | Cefazolina Max Vision | Cefazolina northia | Cefazolina pharmavial | Cefazolina richet | Zolcef;
- (AT) Austria: Cefazolin aptapharma | Cefazolin mip | Cefazolin sandoz | Kefzol | Zolicef;
- (AU) Australia: Cefazolin | Cefazolin aft | Cephazolin alphapharm | Cephazolin sodium | Cepha-zolin viatris | Kefzol;
- (BD) Bangladesh: Cefalin | Cefazol | Zolibac;
- (BE) Belgium: Cefacidal | Cefazoline Mylan | Cefazoline Sandoz | Kefzol;
- (BG) Bulgaria: Cefamezin | Cefazolin | Cefazolin abr | Zepilen;
- (BO) Bolivia, Plurinational State of: Maxibiotic;
- (BR) Brazil: Cefazolina sodica | Ceftrat | Celozina | Cezolin | Fazolon | Kefalin | Kefazol | Zolidina;
- (CH) Switzerland: Cefazolin labatec | Kefzol;
- (CL) Chile: Cefamezin | Cefazolina | Zelaster;
- (CN) China: Cefamezin | Cefamezin sodium | Cefazolin | Cefrazolin sodium | Na lin | Sai fu ning | Tou bao zuo lin na | Xin tai lin | Zhuo zheng;
- (CO) Colombia: Cefacidal | Cefazolina | Cefazolina sodica | Fazopira | Kefzol | Zolibios;
- (CR) Costa Rica: Cefazolina normon;
- (CZ) Czech Republic: Azepo | Cefazolin | Cefazolin noridem | Cefazoline | Kefzol | Orizolin | Vulmizolin;
- (DE) Germany: Baktozil | Basocef | Cefazolin | Cefazolin Eberth | Cefazolin normon | Cefazolin qilu | Cefazolin saar | Cefodin | Cephazolin | Saazolin;
- (DO) Dominican Republic: Alcovera | Cefazolina | Intrazolina | Kefzol | Prisolin;
- (EC) Ecuador: Cefacidal | Cefazolina | Kefzol | Zolcef | Zolidina;
- (EE) Estonia: Cefamezin | Cefazolin | Cefazolin nycomed | Cezolin | Kefzol | Totacef | Zepilen;
- (EG) Egypt: Cefazolin | Kefzol | Totacef | Zinol;
- (ES) Spain: Areuzolin | Brizolina | Cefamezin | Cefaresan | Cefazolina cepa | Cefazolina G.E.S. | Cefazolina ldp-laboratorios torlan | Cefazolina llorente | Cefazolina normon | Cefazolina qilu | Cefazolina Reig Jofre | Fazoplex | Intrazolina | Kefol | Tasep | Tecfazolina | Zepilen | Zolival;
- (ET) Ethiopia: Cefazolin sandoz;
- (FI) Finland: Cefazolin | Kefzol;
- (FR) France: Cefacidal | Cefazoline Flavelab | Cefazoline merck | Cefazoline panpharma | Kefzol;
- (GB) United Kingdom: Kefzol | Trav cepha-zolin;
- (GR) Greece: Kefzol | Vifazolin;
- (HK) Hong Kong: Cefamezin | Cefazolin | Stazolin;
- (HR) Croatia: Cefazolin alpha medical | Cefazolin BCPP | Zepilen;
- (HU) Hungary: Cefazolin aptapharma | Cefazolin human | Cefazolin mip | Cefazolin panpharma | Kefzol | Totacef;
- (ID) Indonesia: Biozolin | Cefamezin | Cefazol | Cefazolin | Evalin | Venozol;
- (IE) Ireland: Cefazolin;
- (IL) Israel: Cefamezin | Cefazolin vit | Kefazin | Totacef;
- (IN) India: Alcizon | Azolin | Cefagood | Cefazolin | Cezolin | Nostof | Orizolin | Prezolin | Prilex | Reflin;
- (IT) Italy: Cefabiozim | Cefamezin | Cefazil | Cefazina | Cefazolina | Cefazolina acs | Cefazolina drm | Cefazolina k24 | Cefazolina union health | Cromezin | Nefazol | Recef | Silzolin | Totacef;
- (JO) Jordan: Cefamezin | Cefazolin sandoz | Hikma cefazolin | Zolecef;
- (JP) Japan: Cefamezin | Cefamezin alpha | Cefamezol | Cefazolin na chm | Cefazolin sodium isei | Cefazopen | Cenazine | Efunikol | Rasenazolin hexal | Rasenazolin maruko | Sefmazon | Synclomedin | Taicezolin sanken | Taicezolin taiyo | Taicezolin towa | Tokio;
- (KE) Kenya: Cefzole | Cezol | Fazlin | Fezolin;
- (KR) Korea, Republic of: Cefamezin | Cefanzol | Cefarizon | Cefazolin | Cefazoline yungjin | Cefozol

- | Ckd cefazolin | Hanlim cefazoline sodium | Kepodon | Kukje cefazoline | Union cefazoline | Yoo young cefazolin sodium | Yuhan cefazoline;
- (KW) Kuwait: Cefazolin sandoz | Zolecin;
- (LB) Lebanon: Cefazoline;
- (LT) Lithuania: Cefamezin | Cefazolin | Cefazolin mip | Cezolin | Intrazoline | Kefzol | Lysolin | Orizolin | Pan-cefazolin | Reflin | Rezol | Totacef | Vulmizolin | Zepilen;
- (LU) Luxembourg: Cefacidal | Cephazolin | Kefzol;
- (LV) Latvia: Cefamezin | Cefazolin | Cefazolin BCPP | Cefazolin mip | Cezolin | Kefzol | Lysolin | Orizolin | Pan-cefazolin | Reflin | Totacef | Vulmizolin;
- (MA) Morocco: Kefzol;
- (MX) Mexico: Cefacidal;
- (MY) Malaysia: Cefazolin | Cefazolin aft;
- (NL) Netherlands: Cefacidal | Cefazoline Aurobindo | Cefazoline Hospira | Cefazoline Mylan | Cefazoline Sandoz | Kefzol | Servazolin;
- (NO) Norway: Cefazolin | Cefazolin hexal | Cefazolin Reig Jofre;
- (NZ) New Zealand: Cefazolin | Cephazolin | Cephazolin sodium | Kefzol | M cefazolin;
- (PA) Panama: Cefazolina normon;
- (PE) Peru: Cekan | Cefacidal | Cefazaliph | Cefazolina | Cephlin | Kefzol | Primacef | Proxicaf | Reflin;
- (PH) Philippines: 2 World Cefazolin | Cefalin | Cefavex | Cefazol | Cefazole | Cefazovit | Cifoxim | Cizo | Cloviz | Faxilen | Fazef | Fazol | Fazolin | Fazonil | Fonvicol | Hazolin | Ilozef | Lozan | May cefazolin | Megacef | Novazef | Phizolin | Samarial | Sanzol | Solosprin | Stancef | Zofacef | Zolival;
- (PK) Pakistan: Afazole | Amrozolin | Avezol | Biozolin | C-zol | Cefamezin | Cefasol | Cefazol | Cefazolin | Cefzo | Celex | Cezolin | Curolin | Kavelin | Kefone | Kefzol | Kefzomed | Medizol | Nelzim | Parmazolin | Pefzol | Sazolene | Zolocef;
- (PL) Poland: Biofazolin | Cefamezin | Cefazolin | Cefazolin aptapharma | Cefazolin mip | Cefazolin tzf | Kefzol | Tarfazolin | Vulmizolin;
- (PR) Puerto Rico: Ancef | Cefazolin | Kefzol;
- (PT) Portugal: Cefazolina hikma | Cefazolina normon | Cefazolina pharmis | Cefazolina sandoz;
- (PY) Paraguay: Cefazoland | Cefazolina genfar | Cefazolina hospicenter | Cefazolina interlabo | Cefazolina millet | Cefazolina northia | Cefazolina richet | Lexibac evim | Mediforina | Novo cefacilin | Novo talexin | Novo thidoxol | Zolidina;
- (QA) Qatar: Cefamezin | Hikma Cefazolin | Zepilen;
- (RO) Romania: Cefazolina atb | Kefzol | Zolinef;
- (RU) Russian Federation: Ancef | Cefamezin | Cefaxon | Cefazolin | Cefazolin akos | Cefazolin Elfa | Cefazolin sandoz | Cefazolin vatchem | Cefezol | Cefoprid | Cezolin | Intrazoline | Kefotex | Kefzol | Lizolin | Lysolin | Lyzolin | Nacef | Orizolin | Orpin | Prozolin | Reflin | Totacef | Zolfin | Zolin;
- (SA) Saudi Arabia: Cefamezin | Cefazolin | Kefzol | Tazoline | Totacef | Zepilen | Zolecin;
- (SE) Sweden: Cefazolin mip | Celmetin | Kefzol;
- (SG) Singapore: Cefazolin | Cefazoline | Cezolin | Zepilen;
- (SI) Slovenia: Altazolin | Cefamezin | Cefazolin actavis | Cefazolin Apta | Kefzol | Zepilen;
- (SK) Slovakia: Cefamezin | Cefazolin | Cefazolin noriderm | Kefzol | Vulmizolin | Zepilen;
- (SV) El Salvador: Cefazolina vijosa;
- (TH) Thailand: Anbizolin | Cefaben | Cefacidal im/iv | Cefalin | Cefamezin | Cefazillin | Cefazol | Cefazolin | Cefazolin meiji | Cefolin | Cefono | Cefzolin | Fazolin | Pharlin | Zefa m h | Zepilen | Zofalin | Zolicef | Zolimed;
- (TN) Tunisia: Cefamezin | Cefazol | Cefazolin | Cefazoline | Zepilen;
- (TR) Turkey: Befazol | Cefamezin | Cefozin | Cezol | Eqizolin | Iespor | Ilksef | Maksiporin | Maxicilin | Sefamax | Sefazol | Vansef;
- (TW) Taiwan: Cefa | Cefacin | Cefamezin | Cefazin | Cefazo | Cefazolin | Cepha | Cephazolin | Cezolin | Kefzol | Kelin | Kofatol | Leadercef | Lofalin | Lyo Sefa | Meisin | Oricef | Sezolin | Stazolin | Uzolin | Veterin | Winzolin;
- (UA) Ukraine: Cefazex | Cefazolin | Cefazolin combi | Cefazolinum | Cezolin | Ifizol | Kefzol | Nacef | Reflin | Totacef | Zolfin;
- (UG) Uganda: Zepilen;

- (UY) Uruguay: Cefazolina | Cefazolina northia | Cefozol | Zolcef | Zolidina;
- (VE) Venezuela, Bolivarian Republic of: Cefacidal | Cefazolina | Cefazomede | Cellozina | Efaztas | Kefzol;
- (VN) Viet Nam: Cefazolin acs dobfar | Erabru | Philfazolin | Shinzolin | Yuhan Cefazolin | Zolifast;
- (ZA) South Africa: Cefacidal | Cefazolin oethmaan | Cefazolin-fresenius | Kefzol | Mylan Cefazolin | Pharmacare cefazolin | Ranzol | Sabax cefazolin | Zefkol;
- (ZM) Zambia: Kefzol

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